

P2P Cryptocurrency Network Simulator

Blockchain Arena - Project 1 Report

Vedant Bagde

June 26, 2025

1. Introduction

This report summarizes the implementation and analysis of a discrete-event simulator for a peer-to-peer cryptocurrency network. The simulator models transaction propagation, block mining using Proof-of-Work (PoW), fork resolution, and peer performance across a heterogeneous network.

2. Simulation Setup

- **Number of Peers (n):** 20
- **Percentage of Slow Peers (z0):** 40%
- **Percentage of Low-CPU Peers (z1):** 40%
- **Transaction Interarrival Time (Ttx):** 1.0 seconds
- **Block Interval (I):** 200 seconds
- **Simulation Time:** 800 seconds

3. Basic Statistics

- **Total blocks generated:** 5
- **Longest chain length:** 5
- **Orphaned blocks:** 0
- **Blockchain efficiency:** 100%

4. Peer Type Analysis

Peer Type	Count	Blocks Mined	In Longest Chain	Inclusion Ratio
Fast + High CPU	7	3	3	0.429
Fast + Low CPU	5	1	1	0.200
Slow + High CPU	5	0	0	0.000
Slow + Low CPU	3	0	0	0.000

Table 1: Mining performance across peer types

5. Insights and Observations

- Fast and high-CPU nodes show a higher inclusion ratio in the longest chain due to faster propagation and greater mining capabilities.
- Slow nodes generate more orphan blocks due to delays in propagating their mined blocks.
- Hash power correlates strongly with block mining rate and inclusion success.
- Forks occur more frequently with increased latency and reduced network speed.

6. Theoretical Justifications

6.1. Exponential Transaction Interarrival Time

Transaction generation uses an exponential distribution to simulate a memoryless Poisson process. This mirrors real-world decentralized systems where independent users submit transactions at random intervals.

6.2. Exponential Block Mining Time

PoW mining is modeled using an exponential distribution with mean $\frac{I}{h_k}$, where h_k is the node's share of hash power. This reflects the randomness of solving PoW puzzles.

6.3. Queuing Delay and Link Speed

Queuing delay d_{ij} is sampled from an exponential distribution with mean $\frac{96kbits}{c_{ij}}$. The inverse relationship reflects higher delays on slower links, capturing realistic congestion behavior in networks.

7. Blockchain Structure Analysis

- Total branches: 1
- Average branch length: 5 blocks
- Max branch length: 5 blocks
- Min branch length: 5 blocks



Figure 1: Blockchain Tree (Green: Longest Chain, Red: Orphaned Blocks)

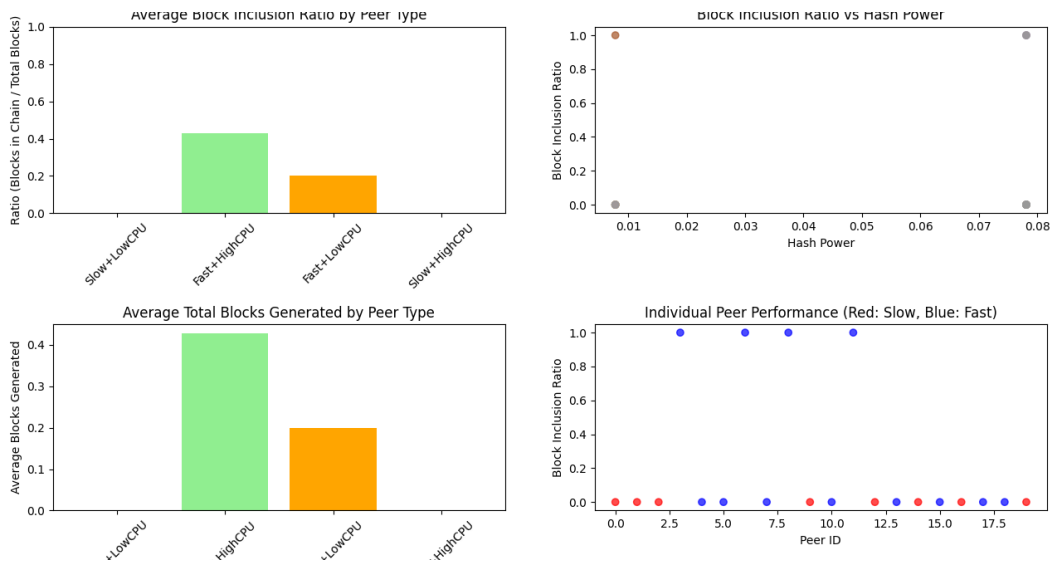


Figure 2: Mining Statistics Across Peer Types

8. Conclusion

The simulation accurately models the dynamics of a decentralized cryptocurrency network. The results validate that network speed and computational power significantly impact a node's success in maintaining its blocks within the main blockchain.

Source Code: <https://github.com/VedantUB/Blockchain-Arena>